ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

CHEMISTRY 6031/3
PAPER 3 2 hours 30 minutes

JUNE 2026 SESSION

Additional materials:
Answer paper
Data Booklet
Electronic calculator

INSTRUCTIONS TO CANDIDATES

Answer **six** questions.

Write your answers on the separate answer paper provided.

If you use more than one sheet of paper, fasten the sheets together.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

This question paper consists of 12 printed pages.

Section A

*Answer any **two** questions from this section.*

1 **(a)** **(i)** Define the term *bond energy*. [1]

(ii) Ammonia reacts with oxygen according to the equation shown.

$$4NH_{3(g)} + 5O_{2(g)} \longrightarrow 4NO_{(g)} + 6H_2O_{(l)}$$

Calculate the enthalpy change of the reaction, given that $N = O$ bond energy is 608 $kJmol^{-1}$.

[use of relevant data from the data booklet is recommended] [3]

(iii) Draw an energy profile diagram for the reaction. [1]

(iv) Explain the relationship between bond length and bond energy in a covalent compound. [2]

(b) **(i)** Carbon monoxide reacts with hydrogen to produce methanol.

Write a balanced chemical equation for the reaction. [2]

(ii) **Table 1.1** shows the standard enthalpies of combustion, ΔH_C^θ, of carbon monoxide, hydrogen and methanol.

Table 1.1

substance	ΔH_C^θ / kJmol^{-1}
carbon monoxide	- 283
hydrogen	- 286
methanol	-715

Calculate the enthalpy change of the reaction in **(i)**. [3]

(c) A mass of 5.00 g of a salt, $Al_2(SO_4)_3 \cdot n\, H_2O$, was heated until the mass was constant at 3.2 g.

Calculate the value of n. [3]

2 (a) (i) Distinguish between

1. *bond length* and *bond polarity*.

2. *dative bond* and *hydrogen bonding*.

[4]

(b) Explain each of the following observations:

(i) a white solid is produced when methylamine and hydrogen chloride gases are mixed;

(ii) NF_3 does not react with water;

(iii) the boiling point of SnH_4 is –52 °C whereas that of SiH_4 is $-112°C$;

(iv) 2,3-dihydroxymethylbenzene has a lower melting point than 2,5-dihydroxymethylbenzene;

(v) graphite with a melting point of about 400 °C is not used as a furnace lining

[7]

(c) (i) State any **two** properties of ceramics. [2]

(ii) Relate each property in **(c)(i)** to the uses of ceramics.

[2]

3 **(a)** Sodium thiosulphate disproportionates in acid solution according to the equation shown.

$$S_2O^{2-}_{3(aq)} + 2H^+_{(aq)} \longrightarrow SO_{2(g)} + S_{(s)} + H_2O_{(l)}$$

Table 3.1 shows the results of the kinetics of the reaction.

Table 3.1

experiment number	$[HCl]$/moldm^{-3}	$[S_2O_3^{2-}]$/moldm^{-3}	time taken
1	0.1	0.01	63
2	0.2	0.01	32
3	0.1	0.05	13

(i) Suggest any **two** methods of determining the rate of the reaction. [2]

(ii) Determine the rate equation. [3]

y:

$32 = k(x)^{0}[y]^{1}$

$k[0.1]^{1}[0.01]$

$(HCl)^{x}$ first order

sec.

(b) A volume of 25.00 cm^3 of a saturated solution of gypsum, $CaSO_4.2H_2O$, was run through an ion exchange resin. A reasonable quantity of water was added to the washings. The wash required 35.00 cm^3 of 0.1 $moldm^{-3}$ $NaOH$. The chemical reactions that occured are as shown in the equations.

$$CaSO_4 + H_2 - Re\sin \longrightarrow Ca - Re\sin + H_2SO_4$$

$$2NaOH + H_2SO_4 \longrightarrow Na_2SO_4 + 2H_2O$$

Calculate the number of moles of

1. $NaOH$ used,
2. $CaSO_4$ in 25.00 cm^3 of the saturated solution.

[5]

(c) **Table 3.2** shows lattice enthalpies of some ionic compounds.

Table 3.2

ionic compound	lattice enthalpy
NaF	- 918 $kJmol^{-1}$
$NaCl$	-771 $kJmol^{-1}$
MgO	-3,79 $kJmol^{-1}$

Explain the differences in the lattice enthalpy values of

1. NaF and $NaCl$,
2. $NaCl$ and MgO.

[5]

le ∝ $\frac{\text{charge}}{\text{radius}}$

[Turn over

Section B

*Answer any **one** question from this section.*

4 **(a)** **(i)** Describe bonding and structure of $AlCl_3$. [2]

(ii) Describe how each of the listed compounds react with water.

1. $NaCl$

2. $MgCl_2$

3. $AlCl_3$

[3]

(iii) Predict the pH of the solution formed when $MgCl_2$ reacts with water. [2]

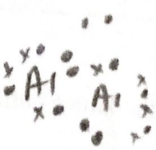
(b) Describe and explain the trend in thermal stability of Group (IV) tetrachlorides. [2]

(c) **(i)** State any **two** uses of Sn. [2]

(ii) Describe the acid-base nature of GeO_2. [2]

(iii) Explain the difference in melting points of Ge and Sn. [2]

energy to break



5 **(a)** Explain each of the following observations:

1. barium sulphate occurs mainly as a solid ore, barite, while magnesium sulphate occurs mainly in solution

2. the decomposition temperature of Group (II) nitrates increases from magnesium to barium

3. barium sulphate is used as a barium meal in taking X-rays of the alimentary canal whilst other compounds of barium are not suitable for this purpose

[4]

(b) An excess of aqueous potassium iodide was added to a portion of 25.00 cm^3 of aqueous chlorine. The iodine liberated required 9.50 cm^3 of 0.120 $moldm^{-3}$ solution of sodium thiosulphate.

(i) Write balanced chemical equations for the reactions that occurred. [2]

(ii) Calculate the number of moles of iodine liberated. [2]

(c) Explain the differences in the reactions of chlorine and iodine with aqueous sodium thiosulphate. [2]

(d) Sulphur dioxide is converted to sulphur trioxide according to the equation shown

$$2SO_2 + O_2 \rightleftharpoons 2SO_3 \qquad \Delta H_r = -191\ kJmol^{-1}$$

(i) Explain why the highest yields of sulphur trioxide are obtained when low temperatures and high pressures are used. [2]

(ii) Describe how the industrial production of sulphur trioxide is made more efficient. [3]

Section C

Answer any two questions from this section

6 Fig 6.1 shows some of the reactions of the organic compound **A**

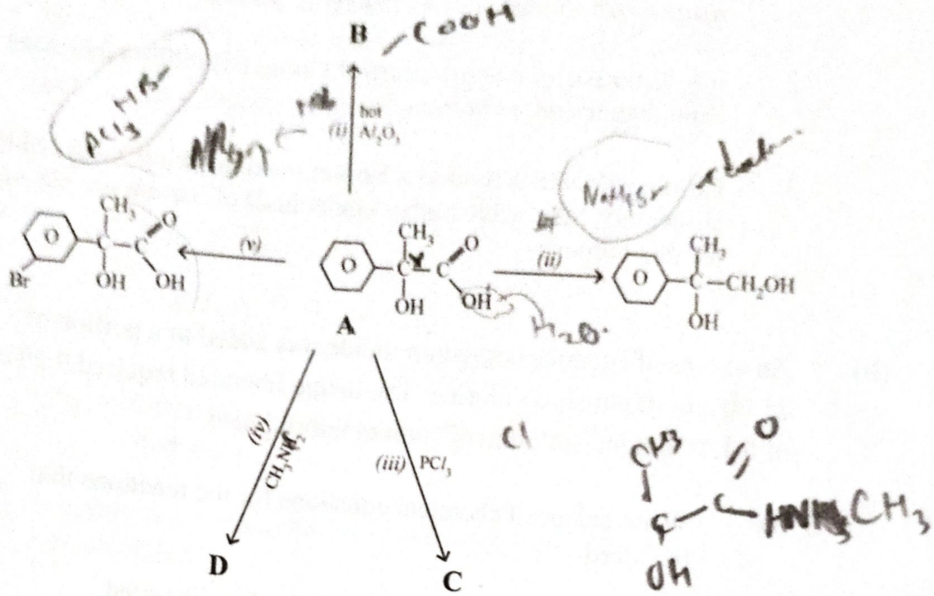


Fig 6.1

(a) **(i)** Name compound **A**. [1]

(ii) State, with a reason, the type of isomerism exhibited by compound **A** [2]

(iii) Comment on the solubility of compound **A** in water. [2]

(b) **(i)** Draw the structural formulae of:

1. **B**,

2. **C**,

3. **D**. [3]

(ii) Give the reagent(s) and condition(s) for reaction:

1. *(ii)*,

2. *(v)*. [5]

(iii) State the observable changes that occur in reaction:

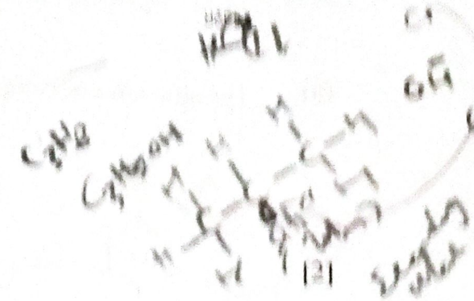
1. *(iii)*,

2. *(v)*. [2]

7 (a) (i) Define the term:

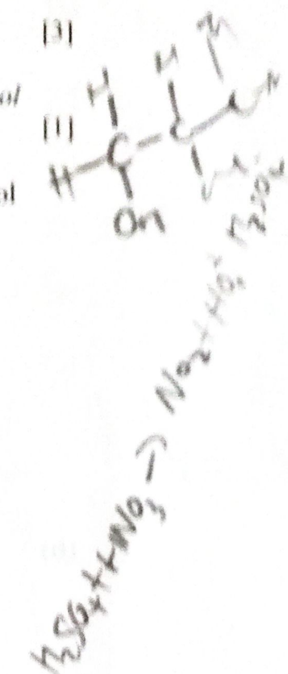
1. *reaction mechanism,*

2. *substitution reaction.* [2]



(ii) Describe the reaction mechanism for the reaction between propan-2-*ol* and hot HCl. [3]

(iii) Explain why the mechanism in **a(ii)** is favoured by propan-2-*ol* but not propan-1-*ol*. [1]



(b) (i) **Fig7.1** shows the reaction scheme for the production of phenol

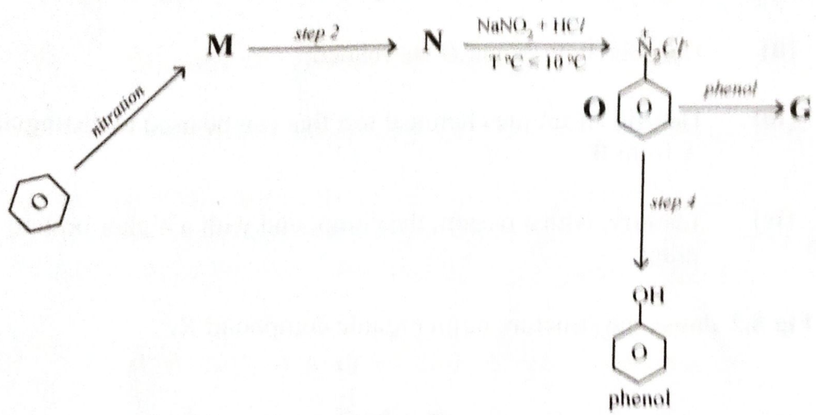


Fig 7.1

Draw structures of

1. **M**,

2. **N**. [2]

(ii) State reagents and conditions

1. for the nitration,

2. *step 2.* [3]

(iii) Write the equation for the reaction in *step 4.* [1]

(iv) State the observation made in *step 4.* [1]

(v) Describe the industrial uses of compound **G** [2]

8 (a) The structures of compounds **A** and **B** are shown in **Fig 8.1**

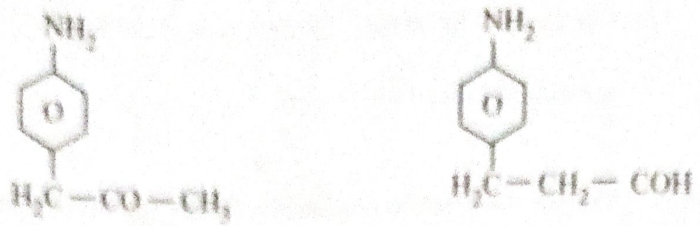


Fig 8.1

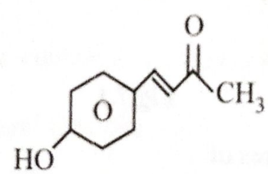
(i) Describe and explain the effect of the amino group on the reactivity of the benzene ring. [3]

(ii) Describe how **A** and **B** are related. [1]

(iii) Describe a simple chemical test that can be used to distinguish **A** from **B**. [3]

(iv) Identify, with a reason, the compound with a higher boiling point. [2]

(b) **Fig 8.2** shows the structure of an organic compound X.



X

Fig 8.2

(i) Name the functional groups present in X. [3]

(ii) Draw the structure of the organic compound formed when X reacts with

1. 2,4 -DNPH,
2. HCN in trace NaOH,
3. cold $KMnO_4$.

[3]

Section D

*Answer any **one** question from this section.*

$x = 0 \quad -2$

Cu_2O

$+1$

$2CuO$

$CuSO_4 + Cu + H_2O$

9 **(a)** Copper (I) oxide disproportionates in aqueous solutions.

(i) Define the term *disproportionation*. [1]

(ii) Write a balanced chemical equation to show the disproportionation of copper (I) oxide in $H_2SO_{4(aq)}$. [1]

(iii) State the observation made in **a(ii)**. [2]

(b) Describe and explain the purification of copper using electrolysis. [3]

(c) **(i)** Give any **two** examples of nanoparticles. [2]

(ii) Describe how nanoparticles can be used in water purification. [3]

(d) A solution of butanedioic acid in ether contained 0.034 moles of butanedioic acid in 20 cm^3 of ether. The solution was shaken with 500 cm^3 of water. The water layer contained 0.032 moles after shaking.

(i) Calculate the concentration of butanedioic acid in the

1. water layer,
2. ether layer.

[2]

(ii) Deduce the partition coefficient. [1]

ode ... oxidation is loss

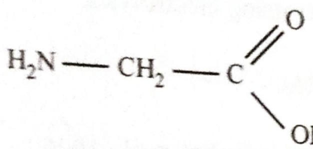
10 (a) (i) Define the term *ligand*. [1]

(ii) Describe and explain the observable changes that occur when aqueous ammonia is added to a solution of cobalt (II) ions and the resultant mixture is left in air. [4]

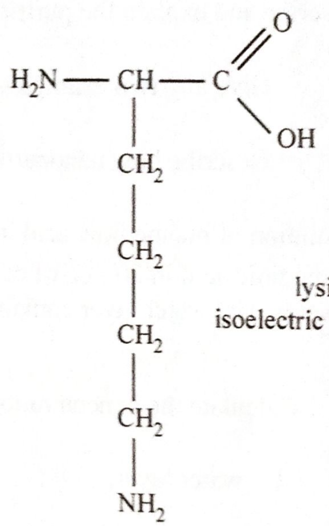
(b) (i) State any **two** methods used to clear up oil spillages at aquatic environments. [2]

(ii) Suggest any **three** causes of oil spillages on land environments. [3]

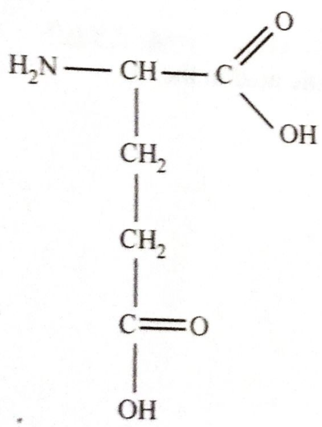
(c) **Fig 10.1** shows the structural formulae of three amino acids and their respective isoelectric points in a buffered mixture at pH 7.



glycine
isoelectric point 5.97



lysine
isoelectric point 9.94



glutamic
isoelectric point 6.90

Fig 10.1

(i) State the charge on each of the amino acids in the mixture. [3]

(ii) Explain how the mixture of the amino acids can be separated. [2]

812.